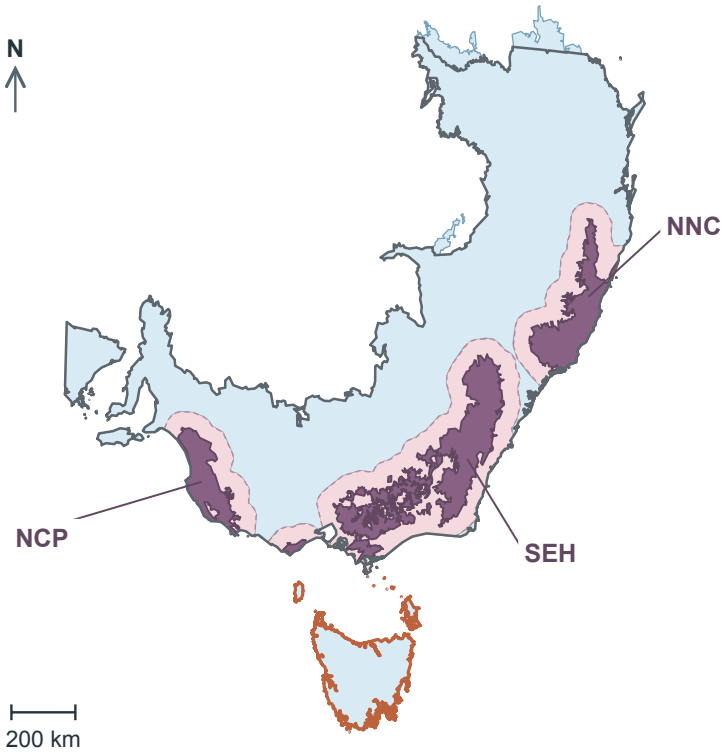
(a) Three geographic tests

Public boundaries in southeastern Australia



Birds: 73 primary species

- Development geography (Tasmania included)
- Three withheld bioregions
- 50 km buffers excluded from all bird fitting

NCP Naracoorte Coastal Plain
SEH South Eastern Highlands
NNC NSW North Coast

Mammals: two further tests

- 5 mainland species → Tasmania (orange outline)
- 8 Tasmanian species tested in withheld patches

Five-patch design: schematic only

Invented positions; no observation locations

a: Boundaries show study extents, not the availability of records or predictors.

(b) Three learners separate two sources of improvement

Pretrained network	Single-species network	Conventional model
Learn from other species (target excluded) → Fine-tune on target species	Fit on target species only Same architecture as the pretrained network	Fit on target species Select within development data only
Predict target distribution	Predict target distribution	Predict target distribution

Pretrained − single-species
What learning from other species adds

Single-species − conventional
What the network adds by itself

Withheld geography is excluded from every stage of fitting and selection.

Additional controls → Supporting information: frozen encoder; scalar community models